Jacobians Part 2

EGR 445/545

GVSU

Plan for Today

- Jacobian review
- Two link Jacobian
- Three link with twist
- Three link planar

Conceptual Review

- what is a Jacobian?
- what is a singularity?
    - mathematically
    - physically

How to Find the Jacobian

- find $\bar{v}_{tip}$ and $\bar{\omega}_N$ symbolically using `sympy` and velocity propagation
- take partial derivatives to find $\mathbf{J}$
- find singularity conditions based on $\|\mathbf{J}\| = 0$

Velocity Propagation

Conceptually:

- we work from the base out to the tip
- velocity of the end of a link is velocity at the base of the link plus the rotational part:
  - $\bar{v}_{tip} = \bar{v}_{base} + \bar{\omega} \times \bar{r}$
    - $\bar{\omega}$ is the total rotational velocity of the link, including rotation of all lower numbered joints
    - $\bar{r}$ is the position vector of the tip with respect to the base of the link
    - vectors for cross products must be in the same frame

Velocity Propagation Equations

- eqn. 5.46: ${}^{i}v_{i+1} \;=\; {}^{i}v_i +^{i} \bar{\omega}_i \times^{i} P_{i+1}$
  - example: ${}^{1}v_2 \;=\; {}^{1}v_1 +^{1} \bar{\omega}_1 \times^{1} P_2$
- eqn. 5.47K: ${}^{i+1}v_{i+1} \;=\; {}^{i+1}_{i}\mathbf{R} \cdot^{i} v_{i+1}$
  - example: ${}^{2}v_2 \;=\; {}^{2}_{1}\mathbf{R}\left({}^{1}v_1 +^{1} \bar{\omega}_1 \times^{1} P_2\right) \;=\; {}^{2}_{1}\mathbf{R}\,{}^{1}v_2$
- eqn. 5.45: ${}^{i+1}\bar{\omega}_{i+1} \;=\; {}^{i+1}_{i}\mathbf{R}\,{}^{i}\bar{\omega}_i + \dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1}$

Two-Link Planar Jacobian

- two link with arbitrary angles
- how do we find the singularity condition(s)?
- walk through all the steps using `sympy`

two_link_Jacobian_full_example.ipynb

Three Link with Twist

- add third link to two-link planar, but $\alpha_2 = 90^\circ$

three_link_with_twist_Jacobian_example.ipynb

Jacobian Learning Activities

Note: check on these

- three-link planar Jacobian